

JAYANTH DASAMANTHARAO

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WORK EXPERIENCE

AI Engineer - Fedway Associates, EliteUS Software Solutions

July 2024 - Present | Edison, NJ

Developing AI systems that connect multiple agents and deliver real-time insights through **chatbots**, User Interface.

- Designed and deployed automated multi-agent AI systems using **AutoGen/Agentic AI**, integrating LLMs with ChromaDB/Azure Search and REST APIs to orchestrate workflows and enable dynamic multi-cloud interactions.
- Built **RAG** pipelines with Azure OpenAI, Amazon Titan, and LLaMA, integrating AI-driven screen scraping to extract and relay live web data into chatbots, improving contextual accuracy, user access to real-time information.
- Implemented advanced **MLOps** practices by automating **CI/CD** workflows for model deployment, versioning, and vector database updates across AWS and Azure, ensuring seamless monitoring and reliability in production.
- Developed **full-stack** AI-powered solutions, including a video/image moderation tool (AWS Rekognition, S3) and an AI chat interface (React), enabling real-time content analysis, human-AI escalation, dynamic UI updates.
- Engineered advanced **LLM evaluation** systems by designing custom metrics (DeepEval) to assess LangChain RAG pipelines, developing semantic matching methods to measure retrieval accuracy and reduce noise.

Data Science Intern, Harvest Software Solutions

June 2023 - Sept 2023 | Jacksonville, Florida

Worked on a digital advertising project, helping advertisers, publishers, and content creators by forecasting ad pricing (CPM) and analyzing customer sentiment to improve ad targeting and **revenue generation**.

- Built CPM **forecasting models** using XGBoost and H2O AutoML, predicting daily & monthly CPM values for advertisers and publishers with an MSE of 0.0321, improving bid optimization and ad spend allocation.
- Engineered large-scale **data pipelines** for ingestion, cleaning, and feature engineering of datasets from Aniview (5M+ rows, 2GB+ files), including outlier handling, normalization, and time-series feature creation.
- Developed NLP pipelines with Hugging Face Transformers and deployed them via FastAPI, performing **sentiment/topic analysis** on millions of records to extract actionable insights for advertisers and publishers.
- Applied **model evaluation and validation** techniques including cross-validation and metrics like MSE, precision, recall, and F1-score, ensuring reliable performance of forecasting and NLP models in production.
- Automated **model deployment**, monitoring through API endpoints and integrated workflows, enabling real-time CPM predictions and insights that boosted accuracy by 70% and reduced manual reporting overhead.
- Machine Learning, Time Series, Data Engineering, SQL, Sentiment Analysis, Deployment & APIs.**

Database Analyst, Accenture Solutions Pvt. Ltd.

June 2021 - Aug 2022 | Hyderabad, India

Client Data Analysis & Data Warehousing, Regeneron Pharmaceuticals Inc.

Managed large healthcare data systems by automating pipelines, maintaining cloud **data warehouses**, and creating dashboards.

- Designed and optimized ETL workflows using Tableau Prep and Python pipelines, streamlining ingestion, transformation, and loading of 30M+ rows of patient and operational data into cloud based warehouse (AWS Redshift and Athena).
- Automated and hardened ETL processes by developing a fail-safe Python script integrated with SQL Workbench and AWS Glue, enabling pipelines to self-trigger, run independently to ensure continuous data processing and avoid downtime.
- Engineered and maintained data warehousing solutions on AWS (Redshift, Athena, S3), integrating multiple disparate data sources into a unified, optimized repository to support large-scale reporting, advanced analytics, access for business teams.
- Improved data warehouse performance by 25% through schema (DDL) optimization, query tuning, and efficient orchestration of large datasets, while deploying ETL workflows, pipeline updates directly to production servers for real-time operations.
- Created Tableau/Power BI dashboards to present key metrics, deliver real-time insights for informed stakeholder decisions.
- Python, AWS, SQL, Data Warehouse, Schema Design, Dashboards, Production Deployment & Monitoring.**

TECHNICAL SKILLS

Programming: Python, R, Scala, MATLAB. **Libraries:** Pandas, Pytorch, Scikit-learn, Tensorflow, SparkML, OpenCV.
Machine Learning: Ensemble Methods, Classification models, Gridsearching, Clustering, Regression Analysis, LightGBM.
Statistical Analysis: Causal Inference, Clustering, Hypothesis testing, Time-series forecasting, Recommendation Engines.
Database & Cloud: SQL, Data Warehouse, AWS(S3, Glue, RDS), GCP, ETL, BigQuery, MongoDB, Langchain, MLOps.
Data Engineering: Databricks, Azure, SQL Workbench, Apache Spark, PostgreSQL, Hadoop, GCP, Tableau Prep, PowerBI.

RESEARCH EXPERIENCE

Researcher - Image Processing & Deep Learning, *Andhra University* Jan 2021 - May 2021 | Visakhapatnam, India

Built an AI-based self-driving car system using **computer vision and sensor fusion** for lane detection and obstacle recognition.

- * Developed and implemented computer vision pipelines for autonomous navigation, including lane tracking, object detection, and environment perception using **YOLO, Faster R-CNN, and CNN-based models**. [IJRESM: [Research Paper](#).]
- * Integrated multi-sensor data (LiDAR, radar, GPS, and cameras) with **deep learning frameworks** to enable real-time detection of vehicles, pedestrians, and obstacles for improved situational awareness.
- * Built lane detection/tracking **algorithms** using HoughTransform, EdgeDetection, CNNs enhancing accuracy in terrains.
- * Engineered an autonomous driving **simulation system using Unity3D** with a Flask backend to monitor real-time parameters (tilt, angle, speed) and validate performance under diverse driving conditions.
- * Performed large-scale model training and experimentation on Google Colab (GPU) using car simulator, applying **image preprocessing, data augmentation, and depth estimation** to improve model generalization and navigation safety.
- * **Python, TensorFlow, Keras, OpenCV, PyTorch, YOLO, Faster R-CNN, LiDAR, GPS, Unity3D, Flask.**

Research Assistant - Statistics, *Rutgers University - NSF* March 2024 - July 2024 | Jacksonville, Florida

Designed a large-scale data collection and analysis pipeline, scraping over 500,000 lawyer profiles and transforming the raw data into actionable legal workforce insights using **advanced statistical and machine learning techniques**.

- * Engineered a Puppeteer-based **web scraping** system, capturing information of 500K+ lawyer profiles with 99% accuracy.
- * Developed a statistical data quality framework using **confidence intervals** to validate data integrity and inconsistencies.
- * Applied advanced analytics (**regression, time-series, clustering**) to uncover trends in lawyer demographics, forecast license status changes, and map legal service hubs across different states in the country.
- * Built anomaly detection pipeline using **Z-score/IQR** methods to automatically flag outliers, missing data, duplicate values.
- * Integrated external datasets for **multi-variable analysis**, providing better insights on how factors influence legal profession.
- * **Node.js, Python (Pandas, NumPy, SciPy, Statsmodels), Regex, Sampling, Confidence Interval Analysis.**

EDUCATION

Rutgers University, *Masters in Data Science* Sept 2022 - May 2024 | New Brunswick, NJ

Relevant Coursework: Statistics: Statistical Modeling and Computing, Stat Learning; Neural Networks, Machine Learning, Data Structures and Algorithms(Python), Data Wrangling (R), NLP, Data Mining. **Projects Overview:** [Portfolio](#).

Andhra University, *B.Tech in Electrical and Electronics Engineering* June 2017 - June 2021 | Visakhapatnam, India

Relevant Coursework: Python, Data Structures & Algorithms, Database Management, Operating Systems, System Design.

PROJECTS

Bayesian Classification - [Bayesian Networks, PyMC3, Informative Prior Distributions] Developed Bayesian logistic regression models using PyMC3 to predict diabetes status, incorporating informative priors. Evaluated performance with recall and f1-score metrics, comparing outcomes between uniform and normal distribution priors.

Language Identification - [Naive Bayes, Logistic Regression, BERT, GloVe embeddings] Worked on identifying languages using Naive Bayes, Logistic Regression, and DistilBERT, incorporating n-grams analysis achieving 96% accuracy.

Twitter Search Application - [Database Management, Python, Sorting techniques, Search Engine Optimization] Created a user query-responsive tweet retrieval system by integrating Postgres for user data management and MongoDB for tweet storage. Optimized data storage and retrieval efficiency by 35% while implementing caching strategies for system performance.

Auto Insurance Fraud Detection - [Python, Data Visualization, ML models - Random Forests, KNN, SVM, Decision Trees, XGBoost, Regression] Executed diverse ML strategies and time series analysis for auto insurance fraud, prioritizing reduced false negatives. Chose K-Nearest Neighbors (KNN) with an impressive 97% recall.

LEADERSHIP

Vice President of Membership - Toastmasters International (2018–2021) – Analyzed membership trends, implemented retention strategies, and led membership drives to increase participation and inclusivity.

Accenture - Mentored a candidate to enhance technical skills and career growth, led cross-functional teams to manage end-to-end deployments, and resolved conflicts to ensure seamless, efficient project delivery in a collaborative environment.

Chegg Inc. - Served as a Subject Matter Expert at Chegg, demonstrating leadership by guiding students through complex data science and AI concepts, providing clear explanations, and fostering independent problem-solving skills.